

CS202-20APCS1-2-Midterm Lab Exam

General

Duration: 75 minutes

Close-book exam.

Programming language: C++.

Submission name: StudentID_MidtermLabExam_YY.zip

- YY: your self-evaluating grade, in [0, 10], 1 mark for 1 method.

Requirement

Define and implement the following classes. Throw exceptions for exceptional cases. Try and catch in main() function.

Define an abstract class, named List, with the following pure virtual (abstract) methods.

- void add (int x);
 - Appends the specified element to the end of this list.
- void add(int index, int x);
 - Inserts the specified element at the specified position in this list.
- void clear();
 - Removes all of the elements from this list.
- List clone();
 - Performs deep-copy action to create a new list.
- int get(int index);
 - Returns the element at the specified position in this list.
- int remove(int index);

- Removes the element at the specified position in this list
- void set(int index, int element);
 - Replaces the element at the specified position in this list with the specified element
- int size();
 - Returns the number of elements in this list.
- List subList(int fromIndex, int toIndex);
 - Return a new list, containing elements in [fromIndex, toIndex)
- string toString();
 - Return a string represented for the list ("10 20 30 40")

Define a concrete class, named LinkedList, inheriting class List, to implement all above methods.

```
struct Node{
    int data;
    Node* next;
};
class LinkedList{
    Node* head;
    Node* tail;
}
```

In the main() function, write some code to test all above methods.

Notes

2 classes are required, missing one of them will get 0 point.

main() function are required, missing main() function will get 0 point.

Linked list implementation are required, dynamic array or vector is not allowed.